

Quiz 1: Foundations of Large Language Models

CO1 · 2 Marks · 15 Minutes · In-Class

Name: _____ Roll No.: _____ Date: _____

Instructions: Circle the letter of the best answer. Q5 requires a written answer (3–5 lines).

★ INSTRUCTOR COPY — Answer key shown in green. Remove for student distribution.

Q1. In the self-attention mechanism, the scaling factor $1/\sqrt{d_k}$ is used to:

- A) Prevent gradient explosion during backpropagation
 - B) Prevent vanishingly small gradients when dot products grow large ✓**
 - C) Reduce overfitting to training examples
 - D) Decrease memory usage at inference time
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Q2. Constitutional AI (CAI) differs from RLHF primarily because:

- A) CAI requires more human labellers
 - B) CAI does not use reinforcement learning at all
 - C) CAI uses AI-generated feedback based on written principles instead of human preference labels ✓**
 - D) CAI only works with smaller models under 7B parameters
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Q3. Which tokenisation method does Claude primarily use?

- A) Character-level tokenisation
 - B) Word-level tokenisation
 - C) Byte-Pair Encoding (BPE) ✓**
 - D) Unigram language model tokenisation
-

Q4. Setting temperature = 0.0 in Claude's API produces:

- A) Deterministic / greedy output — always the highest probability token ✓**
 - B) Uniform random sampling across the full vocabulary
 - C) Top-p = 0 sampling, blocking all output
 - D) The model refuses to generate any response
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Q5. [Short Answer — 3–5 lines] What is the HHH framework and why is it central to Claude?

Model Answer: HHH stands for Helpful, Harmless, Honest — Anthropic's three core training objectives for Claude. Helpful: Claude must maximally assist users in completing their goals. Harmless: Claude must avoid producing outputs that cause physical, psychological, or societal harm. Honest: Claude must be truthful, calibrated in its uncertainty, and non-deceptive. Together these three properties operationalise Anthropic's mission of building safe and beneficial AI, and are directly encoded into the Constitutional AI training process.

Quiz 2: Prompt Engineering

CO2 · 2 Marks · 15 Minutes · In-Class

Name: _____ Roll No.: _____ Date: _____

Instructions: Circle the letter of the best answer. Q5 requires a written answer (3–5 lines).

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Q1. "Let's think step by step" is the trigger phrase for which prompting technique?

- A) Few-shot prompting
- B) Zero-shot Chain-of-Thought (CoT) ✓**
- C) Self-consistency prompting
- D) ReAct prompting

Q2. Self-consistency improves accuracy by:

- A) Using a larger model for each query
- B) Sampling multiple diverse reasoning chains and selecting the majority answer ✓**
- C) Having the model critique its own output once
- D) Adding more in-context examples

Q3. In the ReAct framework, the "Action" step refers to:

- A) The model's final answer to the user
- B) Human feedback on the model's output
- C) A tool call or external operation triggered by the model ✓**
- D) Rewriting the original prompt

Q4. Which of the following is NOT a direct benefit of structured output prompting?

- A) Easier downstream parsing of model responses
- B) Consistent, predictable response format
- C) Better integration with APIs expecting JSON
- D) Guaranteed elimination of hallucinations ✓**

Q5. [Short Answer] Explain the difference between a system prompt and a user prompt in Claude.

Model Answer: A system prompt is injected by the developer/operator before the conversation begins. It defines Claude's persona, task scope, constraints, tone, and output format. It is not visible to the end-user and frames how Claude should interpret all subsequent messages. A user prompt is the message sent by the end-user during the conversation. System prompts take precedence and act as persistent instructions. Example: a system prompt might say "You are a tax assistant. Only answer tax-related questions." while user prompts contain the actual questions.

Quiz 3: Claude API & Applications

CO3 · 2 Marks · 15 Minutes · In-Class

Name: _____ Roll No.: _____ Date: _____

Instructions: Circle the letter of the best answer. Q5 requires a written answer (3–5 lines).

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Q1. In a RAG pipeline, the vector database is used to:

- A) Store the LLM model weights
 - B) Store and retrieve document embeddings using similarity search ✓**
 - C) Cache API responses to reduce cost
 - D) Store multi-turn conversation history
-

Q2. When Claude's API returns stop_reason = "tool_use", the developer must:

- A) Retry the API call with a higher temperature
 - B) Execute the requested tool and return its result as the next message ✓**
 - C) Increase the max_tokens parameter
 - D) Switch to a different Claude model
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Q3. To pass an image to Claude's Vision API, it must be:

- A) Encoded as UTF-8 text
 - B) Encoded as hexadecimal
 - C) Encoded as base64, or provided as a publicly accessible URL ✓**
 - D) Encoded as raw binary data
-

Q4. Claude's prompt caching feature primarily reduces:

- A) Cost and latency for large, repeated context prefixes ✓**
 - B) Hallucination rate in model responses
 - C) The need for system prompts
 - D) Model size at inference time
-

Q5. [Short Answer] What is wrong with this API call:

messages=[{"role":"user","content":"Hi"},{"role":"user","content":"What can you do?"}]?

Model Answer: The Claude API requires strictly alternating user/assistant turns. Two consecutive messages with role="user" are invalid and will result in an API error (400 Bad Request). The correct pattern is user → assistant → user → assistant. After the first user message "Hi", the developer must include the model's response (role="assistant") before sending the next user message. Fix: insert {"role": "assistant", "content": "Hello! How can I help you?"} between the two user messages.

Quiz 4: Responsible AI & Ethics

CO4 · 2 Marks · 15 Minutes · In-Class

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Instructions: Circle the letter of the best answer. Q5 requires a written answer (3–5 lines).

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Q1. "Extrinsic hallucination" in LLMs refers to:

- A) Internal contradictions within a single model response
 - B) Generated content that cannot be verified from or contradicts the source document ✓**
 - C) Errors caused by low temperature sampling
 - D) Hallucinations in multilingual settings only
-

Q2. Under the EU AI Act, AI systems used in employment screening are classified as:

- A) Minimal risk
 - B) Limited risk
 - C) High risk ✓**
 - D) Unacceptable risk
-

Q3. Which strategy is most effective at grounding Claude's answers to a specific document?

- A) Increasing temperature to 1.5
 - B) Using the word "accurate" in the system prompt
 - C) RAG — injecting retrieved document chunks into the context ✓**
 - D) Using Claude-3-Opus instead of Haiku
-

Q4. A prompt injection attack aims to:

- A) Overload the API server with too many requests
 - B) Override the developer's system prompt by embedding adversarial instructions in user input ✓**
 - C) Steal API keys embedded in API responses
 - D) Reduce the model's effective context window
-

Q5. [Short Answer] What is the WinoBias benchmark and what does it measure?

Model Answer: WinoBias is a benchmark dataset used to evaluate gender bias in NLP systems, specifically in coreference resolution. It contains sentence pairs testing both stereotypical associations (e.g., "the nurse... she") and anti-stereotypical associations (e.g., "the nurse... he"). A system that performs significantly better on stereotypical examples than anti-stereotypical ones exhibits gender bias. High disparity scores indicate the model relies on gender stereotypes rather than genuine linguistic cues.

Quiz 5: Advanced Topics

CO5 · 2 Marks · 15 Minutes · In-Class

Name: _____ Roll No.: _____ Date: _____

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Q1. In the Model Context Protocol (MCP) architecture, an "MCP Server" is responsible for:

- A) Running the Claude model weights locally on the user's machine
 - B) Exposing tools, resources, and capabilities (e.g., file system, database) to the LLM host ✓**
 - C) Managing user authentication and session tokens
 - D) Compressing and batching API requests for efficiency
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Q2. In a multi-agent "orchestrator-worker" pattern, the orchestrator's primary role is:

- A) Directly executing all tool calls
 - B) Planning tasks, delegating to worker agents, and aggregating results ✓**
 - C) Providing the user-facing UI layer
 - D) Monitoring API costs and triggering alerts
-

Q3. RAG is preferred over fine-tuning when:

- A) The knowledge domain is very narrow and completely static
 - B) The knowledge base changes frequently and needs real-time updates ✓**
 - C) Inference latency is not a concern
 - D) Embedding models are not available
-

Q4. Claude's "extended thinking" mode is designed to improve performance on:

- A) Short creative writing tasks
 - B) Image generation
 - C) Complex multi-step reasoning and mathematics problems ✓**
 - D) Real-time streaming applications
-

Q5. [Short Answer] Name THREE strategies to reduce Claude API costs in a production application.

Model Answer: (1) Prompt caching: Cache large, repeated system prompt prefixes so they are only processed once, reducing input token costs on subsequent calls. (2) Model tiering: Route simple queries to claude-3-haiku (cheapest) and reserve claude-3-5-sonnet or Opus for complex tasks that actually require it. (3) Output length control: Set max_tokens appropriately and use concise prompts to avoid unnecessarily long responses. Bonus strategies: Use the Batch API for non-real-time workloads (50% discount); compress or summarise conversation history to reduce context length.